



Is Disruption On The Cards For Passives



Dilin Anand is a senior assistant editor at EFY. He is B. Tech from University of Calicut, currently pursuing MBA from Christ University, Bengaluru

It is one of the most boring sectors in electronics,” said one of the first people interviewed for this story. While most design engineers who were spoken to were of the view that the passive components segment was not seeing any significant advancements in technology, increasing demand from sectors built around buzzwords like the Internet of Things (IoT), 3D printing and space electronics are driving passive component manufacturers to research and develop better product-manufacturing technologies. Decreasing demand for conventional electronic devices enhances this effect, as the new driving factor is for smaller and higher-quality parts with better performance.

According to a report by *Paumanok Publications* on the passive electronic components industry, “Many component manufacturers are now transitioning towards value-added and application-specific end-use market segments that require added tooling and captive intellect.” This means that manufacturers will now need a deeper technical understanding of high-voltage, high-frequency and harsh-environment end-use markets in order for them to come up with competitive solutions.

Passives designed for specific uses

“Many passive components now have extended higher-temperature performance, which previously were suited more for military and aerospace industries. As applications from other industries such as medical and industrial also require certain higher specifications, many manufacturers have extended their product range into these industries,” says Navin Honnavar, senior manager for marketing (South Asia), Premier Farnell.

Components for these industries are now being made less expensive by exploring alternative compounds for their design. An example is the water-insoluble nitride-film based resistor from TT Electronics that can resist moisture-vulnerability problems of nichrome and passivated-nichrome resistor-film technologies. This was previously only possible using high-cost military-grade tantalum-nitride. This moisture resistance means users do not need to rely on expensive assembly-level sealing or encapsulating processes.

Vishay Intertechnology has also come out with resistors based on moisture-resistant tantalum-nitride technology that